MAI 580 Course Project — Milestone 1 Progress Note (Stages 1–5)

Ayman Ketat · Use Case B: Research Assistant · 24 October 2026
Repository: https://github.com/aymanketat/mai580-agent (private; aniskoubaa invited) · tags stage-01 to stage-05
Model gpt-4.1-mini (temperature 0); langgraph 1.2.12, langchain 1.4.2, langchain-openai 1.6.6; Python 3.11.
Supporting evidence: notes/M1_detailed.md (all tables, full reflection answers), eval/ (question sets, raw runs, grades, grader prompts).

1. Implementation

The agent answers questions about machine-learning papers. Its data is a catalogue of 23 arXiv records (data/papers.json, built from the arXiv API) and, from Stage 5, live citation counts from Semantic Scholar.

stage	graph	added
1	START → model → END	one State with messages and the add_messages reducer; call_model node
2	analyze → generate → review	fields analysis, draft; three focused prompts (kept as build_workflow())
3	classify → one of five answer nodes	structured-output classifier over paper_lookup / literature_search / citation / concept / other; conditional edge
4	router + agent → tools → final_answer	three read-only tools: search_papers (lexical, catalogue), get_paper, format_citation (APA/BibTeX/IEEE); errors returned as data
5	router + agent ⇄ tools	loop closed (tools → agent); recursion_limit = 12 with a fallback message; get_citation_count, check_citation

Tests: the suite at each tag passes 2 / 2 / 4 / 7 / 15 tests (re-run from checkouts of the tags, eval/results/test_runs.md). After the post-review fixes the current tree has 65 tests: 60 offline (tools, cache, reference checks, graph structure; no model, no network) and 5 live (model calls), all passing.

Post-review fixes within Stage 5 (correction record in the detailed report): stricter reference verification (only an identical or subtitle-omitted title verifies; near matches are reported as candidates), surname-first author names, a persistent citation-count cache with retrieval times and an offline mode, Retry-After HTTP-dates, and the step budget in the inspection script.

2. Experiments

Stage 1 — five research questions. 2 correct, 2 honest refusals, 1 confident invention (the VAE paper's latest version given as "v10"; the catalogue says v11). Two separate runs of "My name is Sara" and "What is my name?" show no memory between runs.

Stage 2 against Stage 1 on the same 10 questions (three blind LLM graders, median score 1–5):

system	latency	tokens in / out	quality	answers with a factual error
Stage 1, single node	34.7 s	307 / 4 330	3.9	2
Stage 2, workflow	66.4 s	8 257 / 7 998	3.4	2 (plus 3 stale or unsupported additions)

The review node changes tone and length, not facts: it hedged correct answers and accepted "Llama 2 has 13 authors" (the catalogue lists 68). It helped only on the fake-reference check.

Stage 3 — routing accuracy on 20 labelled requests: 18/20 (90%) with the first category descriptions; both misses were reference checks routed to paper_lookup. After rewording the citation description: 20/20. The rewording was tuned on the same 20 items, so 100% is an upper bound.

Stage 4 against Stage 5 on 10 multi-step requests and 1 single-step control (three blind graders; one rubric then applied to both stages, keeping task completion, factual correctness and tool provenance separate):

measure (multi-step requests 1–10)	Stage 4: one round, 3 tools	Stage 5: loop, 5 tools
verified success (all parts delivered, correct, citations and counts traceable to a tool call)	0/10	5/10
unverified (delivered, but a part rests on tool results that were not logged)	0	1 (Q5)
content-only success (provenance not required)	3/10	5/10 (+1 unverified)
answers with a factual error	2	0
answers with an unsupported claim	3	2
answers with a citation lacking tool provenance	4	0
mean score, recorded medians · rubric re-application	2.10 · 2.60	4.00 · 3.60
tool calls · input tokens	13 · 11 980	36 · 44 291
single-step control (request 11)	1/1	1/1

All four Stage 5 failures were Semantic Scholar rate limits, reported as "could not retrieve"; a later re-run with one retry completed three of them (recorded separately). On the five requests that are solvable with Stage 4's tools but need two

dependent calls, Stage 4 scored 0/5 verified (3/5 content-only) and Stage 5 5/5 — evidence consistent with a benefit from the loop, not a controlled isolation of it.

3. Findings

- The model refuses when a question states that it needs live or private data, and invents when the fact looks ordinary but is rare in training text (version numbers, author counts, id-to-title mappings).
- Adding steps without evidence (Stage 2) adds places to be wrong; adding tools (Stage 4) fixes the facts the catalogue holds; closing the loop (Stage 5) makes dependent calls possible.
- The loop costs 2.8× tool calls and 3.7× input tokens (every round re-sends the history and the tool schemas) and wastes calls: 8 unnecessary calls in 11 requests (re-searching an id already in hand, counting off-topic search hits). These motivate Stage 8 (retrieval quality) and Stage 9 (planning).

4. Limitations and open points

- Stage 4 → 5 changes the loop and adds two tools at once; 5 of the 10 multi-step requests need a citation count that only a Stage 5 tool provides, so the comparison measures the combined upgrade.
- The grader prompt wrongly stated that Stage 4 had five tools. Success verdicts are unaffected; an independent re-grade confirmed the Stage 4 scores for Q2, Q3, Q9 and Q10 under the correct tool list. Q6 was reclassified after checking the official NeurIPS 2014 metadata: its conference citation is correct and only tool provenance is missing (content-only success 2/10 → 3/10).
- Q5 (Stage 5): the citation count and ranking are unverified because the run log holds tool calls, not tool results; the request is not counted as a verified success. Logging tool results is a Stage 12 item.
- Q8: CLIP was selected by both systems. It is relevant (the paper trains Vision Transformer image encoders, Section 2.4), but the retrieved abstract does not say so; the selection is counted as correct but unsupported by the agent's evidence.
- Q9 (Stage 4): "The paper is not authored by Yao" is a factual error as written (Yao is the sixth author); the preceding statement that Yao is not the first author was correct, so the answer was internally inconsistent.
- Routing accuracy is measured on the tuning set; Semantic Scholar's unauthenticated API rate-limits readily.

5. Reflections (summary; full answers in the detailed report)

A node returns an update and the reducer merges it; without `add_messages`, a node that returns `messages` replaces the history, while nodes that return only `analysis` or `draft` leave it unchanged. The single edge `tools → agent` turns the workflow into an agent. `recursion_limit` is both a cost and a safety control, but it counts steps rather than cost or risk. A structured-output classifier is safer than a written label because the schema constrains the output.

6. Contribution and AI disclosure

I selected Use Case B and am working through the code and results. Claude Code (Anthropic, model `claude-fable-5-1`) assisted with implementation, testing and documentation: it drafted the code for each stage from the tutorial's starter code, wrote the experiment scripts and question sets, ran the test suites and experiments, and drafted this note and the detailed report. The automated checks reported here — `pytest`, the label review of the question sets, and the blind grading of the Stage 2 and Stage 5 comparisons — were performed by Claude instances and are distinct from my own review, which is in progress.